

Nikhil

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PROFILE

AI/ML Engineer and Researcher specializing in Generative AI, LLM applications, RAG, agentic AI, and applied machine learning, with experience building production-grade AI systems and applications across web and mobile platforms. Experienced in developing scalable AI systems using Python, FastAPI, cloud platforms, and modern LLM architectures, with research experience in AI-driven unsupervised learning, and generative modeling.

EXPERIENCE

AI/ML Researcher - Vellore Institute of Technology (Apr 2025 – Jul 2026) - Vellore, India

- Developed an AI system for semantic content extraction and context-preserving video summarization.
- Examined unsupervised learning, generative modeling, sequence architectures, attention mechanisms, and feature representations for video understanding.
- Engineered and evaluated summarization pipelines using benchmark datasets and quantitative evaluation metrics to improve summary quality.

Python Developer - Meta Scifor Technologies (Jan 2024 – Apr 2025) - Bengaluru, India

- Engineered Python-based backend systems and AI-powered chatbot functionality for an E-Learning platform, supporting core learning and application workflows.
- Engineered the Python backend for a competitive-events platform supporting hackathons, coding competitions, registrations, and participant workflows.
- Developed data-driven platforms, including an Athlete Management System for managing athlete data and performance metrics, while delivering scalable backend solutions across multiple web applications.

PROJECTS

AI Live Interview Platform (CrackAI App)

- Built a production-scale AI interview platform using LLMs, AI personalization, and hybrid AI workflows.
- Engineered scalable FastAPI services integrating AI interviews, resume analysis, and intelligent feedback on GCP.
- **Tech Stack:** Gemini AI • FastAPI • Firebase • Firestore • Docker • GCP • Prompt Engineering

Enterprise Knowledge Copilot

- Built an enterprise RAG platform with semantic search, document retrieval, and citation-backed AI responses.
- Engineered retrieval pipelines using embeddings, vector databases, reranking, and LangChain.
- **Tech Stack:** LangChain • LlamaIndex • Pinecone • FastAPI • LangSmith • Ragas • MLflow

Multi-Agent AI Research Assistant

- Built a multi-agent AI system enabling collaborative reasoning, planning, and autonomous task execution.
- Engineered agent orchestration with LangGraph, MCP, tool calling, memory, and scalable FastAPI services.
- **Tech Stack:** LangGraph • MCP • FastAPI • Docker • Kubernetes • Redis

TECHNICAL SKILLS

- **AI & LLM:** LLM Application Development • Prompt Engineering • Structured JSON Outputs • JSON Schema Validation • RAG • Embeddings • Vector Databases • LangChain • LlamaIndex • LangGraph • AI Agents • MCP • Tool Calling • AI Personalization • Ragas/DeepEval • LangSmith
- **Machine Learning:** Python • PyTorch • Transformers • Hugging Face
- **Backend & Cloud:** FastAPI • REST APIs • Docker • Kubernetes • Google Cloud Platform (Cloud Run) • Firebase • Firestore • Redis
- **Software Engineering:** Git • Linux • SQL • System Design • CI/CD • Authentication(JWT/OAuth) • Microservices

EDUCATION

M.Tech. Software Engineering

- Vellore Institute of Technology (2024)